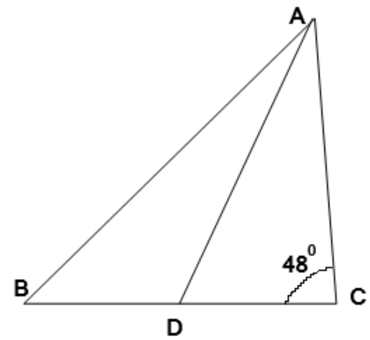
THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
NMTC JUNIOR LEVEL – IX & X Grades PRELIM TEST

NMTC – 2023

1. a, b, c are real numbers such that $x^3 + 6x^2 + ax + b$ is the cube of $(x+c)$ then:
   (a) $(a+b+c)$ is divisible by 13 (b) $a+b=11c$
   (c) $a>b$ and $b<c$ (d) $(a+b+c)$ is divisible by 11.

2. In the adjoining figure, AB = 9 cm, AC = 7 cm, BC – 8 cm. AD is the median and $\angle C = 40^0$. Then measure of $\angle ADB$ (in degrees) is

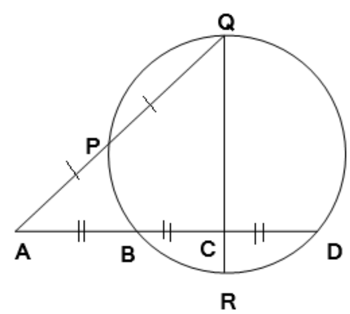


   (a) 100 (b) 140 (c) 45 (d) 120

3. If $x^2+6x+1=0$ and $\dfrac{x^4+kx^2+1}{3x^3+kx^2+3x}=2$, then the value of k is
   (a) 68 (b) 72 (c) 65 (d) 70

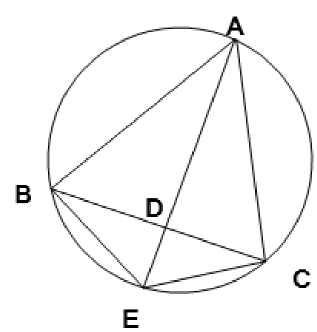
4. If $x=\sqrt[3]{49}+\sqrt[3]{42}+\sqrt[3]{36}$, then the value of $x-\dfrac{1}{x^2}$ is
   (a) $2\sqrt[3]{42}$ (b) $3\sqrt[3]{42}$ (c) $\sqrt[3]{42}$ (d) $4\sqrt[3]{42}$

5. In the adjoining figure, AB = BC = CD. P is the midpoint of AQ. If CR = 4, QC = 12 then PQ is equal to:



   (a) $4\sqrt{3}$ (b) $6\sqrt{3}$ (c) $8\sqrt{3}$ (d) $2\sqrt{3}$

6. In the adjoining figure, A is the midpoint of the arc BAC, Given that AB = 15 and AD = 10. Then the value of AE is:



   (a) 22 (b) 23 (c) 22.5 (d) 23.5

7. The number of real numbers $4\sqrt{3}$ which satisfy the equation $\frac{8^x + 27^x}{12^x + 18^x} = \frac{7}{6}$ is

(a) 1 (b) 2 (c) 0 (d) 4

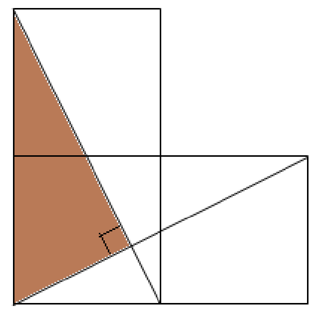
8. a, b are real numbers such that $2a^2 + 5b^2 = 20$. Then maximum value of $a^4 b^6$ is

(a) 256 (b) 1024 (c) 1262 (d) 16

9. The number of order pairs (x, y) of integers such that $x - y^2 = 4$ and $x^2 + y^4 = 26$ is

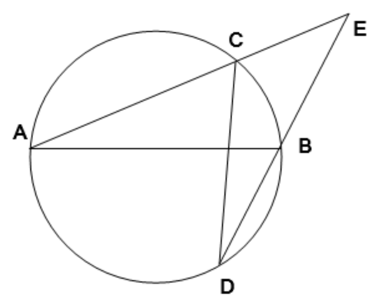
(a) 4 (b) 3 (c) 2 (d) 1

10. In the adjoining figure, three equal squares are placed. The squares are unit squares. The area of the shaded region (in cm^2) is



(a) $\frac{5}{4}$ (b) $\frac{4}{5}$ (c) $\frac{3}{2}$ (d) $\frac{3}{4}$

11. In the adjoining figure, AB is a diameter of the circle. Given $\angle BAC = 20^0$, $\angle AEB = 56^0$. Then the measure (in degrees) of $\angle BCD$ is



(a) 12 (b) 10 (c) 14 (d) 16

12. The number of ordered pairs (m, n) of integers such that $1 \le m, n \le 100$ and $m^n n^m$ leaves a remainder 1 when divided by 4 is:

(a) 2250 (b) 1000 (c) 1125 (d) 1250

13. The number of ordered pairs of positive integers (x, y) satisfying the equation $x^2 + 4y = 3x + 16$ is

(a) 1 (b) 2 (c) 3 (d) 4

14. The algebraic expression $(a + b + ab + 2)^2 + (a - ab + 2 - b)^2 - 2b^2(1 + a^2)$ reduces to:

(a) $4(a+2)^2$ (b) $2(a+2)^2 + 4ab^2$ (c) $(a-2)^2$ (d) $2(a-2)^2 + 4ab^2$

15. The sum of $1\times4 + 2\times7 + 3\times10 + 4\times13 + \ldots$ 49 terms is equal to:

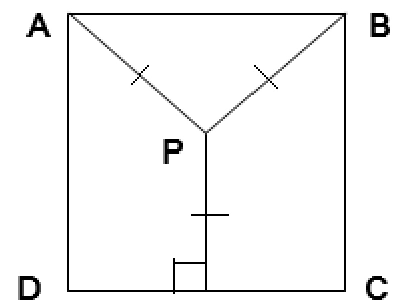
(a) 12250 (b) 116800 (c) 11800 (d) 117600

16. If the equation $x^3 + ax + 1 = 0$ and $x^4 - ax^2 + 1 = 0$ have common root, then the value of a^2 is __________.

17. If a,b,c,d are positive reals such that $abcd = 1$, then the maximum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is _________________.

18. The sum of all natural numbers n for which $n(n+1)$ is a perfect square is __________.

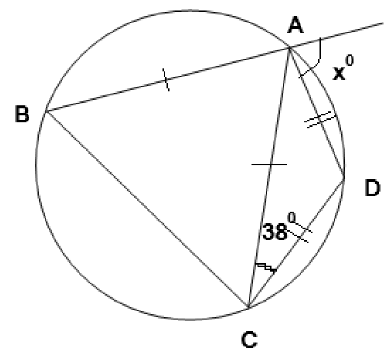
19. P is appoint inside the square ABCD such that PA = PB = Distance of P from CD. The ratio of the areas of the triangle PAB to the area of the square ABCD is $\frac{m}{n}$, where m,n are relatively prime integers. Then value of $m + n =$___________.



20. The sum of roots of the simultaneous equation: $\sqrt[y]{4^x} = 32\sqrt[x]{8^y}$, $\sqrt[y]{3^x} = 3\sqrt[x]{9^{1-y}}$ is ___________.

21. If $2\sqrt{3+\sqrt{5-\sqrt{13+\sqrt{48}}}} = \sqrt{a} + \sqrt{b}$, where a,b are natural numbers, then the value of $a+b$ is ____________.

22. In the adjoining figure, $\angle ACD = 38^0$. Then the measure (in degrees) of angle x is _____________.



23. If $\frac{a}{b+c} = \frac{c}{a+b} = \frac{2b}{c+a}$ (where $a+b,\ b+c,\ c+a,\ a+b+c$ are all not zero), then the numerical value of $\frac{a^2+b^2}{c^2}$ is ______________.

24. The geometric and arithmetic means of two positive numbers are respectively 8 and 17. The larger among the two numbers is _________.

25. The numbers of two-digit numbers in which the tens and the units digit are different and odd is _______________.

26. The value of $\left(5\sqrt[3]{4} - 3\sqrt[3]{\frac{1}{2}}\right)\left(12\sqrt[3]{2} + \sqrt[3]{16} - 2\sqrt[3]{2}\right)$ is equal to ______________.

27. If $\frac{xy}{x+y}=1,\ \frac{yz}{y+z}=2,\ \frac{zx}{z+x}=3$, then the numerical value of $15x-7y-z$ is ____________.

28. The sum of all natural numbers which satisfy the simultaneous inequations $x+3<4+2x$ and $5x-3<4x-1$ is _____________.

29. In an increasing geometric progression (with 1st term *a*, nth term t_n), the difference between the fourth and the first term is 52 and the sum of first three terms is 26. Then numerical value of $\frac{t_{2024}}{t_{2023}}+\frac{a^{2024}}{a^{2023}}$ is ___________.

30. The base of a triangle is 4 units less than the altitude drawn t it. The area of the triangle is 96 (unit^2). The ratio of the base to the height is $\frac{p}{q}$ where p, q are relatively prime to each other. Then value of $p+q$ is_________.

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